Ionic Conductivity Calculations

This tutorial demonstrates how to calculate the ionic conductivity of lithium ions (Li^+) use the `GPUMD` package. We will use the garnet-type $Li_7La_3Zr_2O_{12}$ (LLZO), a well-known solid electrolyte, as our model system. You will learn how to:

- Compute the mean squared displacement (MSD) of Li^+ ions.
- Derive the diffusion coefficient from MSD data.
- Convert the diffusion coefficient to ionic conductivity using the Nernst-Einstein relation.
- Estimate the activation energy of Li^+ diffusion.

Setup and Resources

To follow this tutorial, you will need the following:

- **Input Structure**: `model.xyz` file for the cubic LLZO (c-LLZO).
- **Potential File**: a well-trained NEP (`nep.txt`) for the c-LLZO system.
- **GPUMDkit package**: https://github.com/zhyan0603/GPUMDkit

In this tutorial, I provide the `NEP25-ZBL` model trained in our previous work [1], which is a NEP trained with only 25 configurations and successfully describes various properties of LLZO, including phase transitions, atomic local environments, and ionic diffusion properties. For more details, refer to https://github.com/zhyan0603/SourceFiles and *J. Chem. Inf. Model.* 2026, 66, 3, 1406–1413.

Step 1: Grouping Elements

It is recommended to create a separate directory for each target temperature (e.g., 950K, 1000K, 1100K, and 1200K). Inside each temperature folder, place your specific `run.in` and `model.xyz` files. For better stability and accuracy, the `model.xyz` you provide here should ideally be a **pre-equilibrated** structure rather than a raw, unrelaxed one.

After that, you need to group the elements in your structure. With the latest `GPUMDkit`, you can easily do this using the following command: `gpumdkit.sh -addgroup <inputfile> Li La Zr O`

The `<inputfile>` can be a `POSCAR.vasp`, an `extxyz` file, or any other format readable by the `ase` package. This command will generate a `model.xyz` file where atoms are grouped according to your specified element order.

*(**Note: Make sure your diffusing ion, like Li, is placed first**).*

Step 2: Simulation Setup

Here is an example input file for performing the simulation at 1000K:

```
potential ../nep.txt
velocity    1000

ensemble    npt_mttk temp 1000 1000 aniso 0 0
compute_msd 100 1000 all_groups 0 save_every 100000
dump_thermo 100
dump_restart 1000000
run         1000000
```

Explanation of Input Parameters

- `potential ../nep.txt`: Specifies the NEP file for LLZO.
- `velocity 1000`: Initializes atomic velocities at 1000 K.
- `ensemble npt_mttk temp 1000 1000 aniso 0 0`: Uses the NPT ensemble with the Martyna-Tuckerman-Tobias-Klein thermostat and barostat at 1000 K.
- `compute_msd 100 1000 all_groups 0 save_every 100000`: Calculates the MSD of all species. The position will be recorded every 100 timesteps, and the maximum number of correlation steps is 1000. For `save_every` command, `GPUMD` will write `msd_step*.out` file every 100000 steps. See here for details of `compute_msd` command.
- `dump_thermo 100`: Outputs thermodynamic properties to `thermo.out` every 100 fs.
- `dump_restart 1000000`: Saves the `restart.xyz` file.
- `run 1000000`: Performs a 1 ns production run.

Step 3: Calculating and Visualizing MSD

The MSD quantifies the average distance traveled by Li^+ ions over time, a key quantity for calculating ionic diffusivity. The MSD is calculated using the relation:

$$\mathrm{MSD}(t) = \frac{1}{N}\sum_{i=1}^{N}\left\langle \left|\mathbf{r}_i(t) - \mathbf{r}_i(0)\right|^2 \right\rangle$$

- $\mathbf{r}_i(t)$ is the position of ion (i) at time (t).
- $\mathbf{r}_i(0)$ is the initial position of ion (i).
- N is the number of Li^+ ions.
- $\langle\ \rangle$ denotes the time average.
- $\frac{1}{N}\sum_{i=1}^{N}$ denotes the averages over all Li^+ ions.

After running the simulation, `GPUMD` generates an `msd.out` file containing the MSD data for Li^+ ions. See here for details of `msd.out` file.

Visualizing MSD

The `msd.out` file contains the MSD and self diffusion coefficient (SDC).

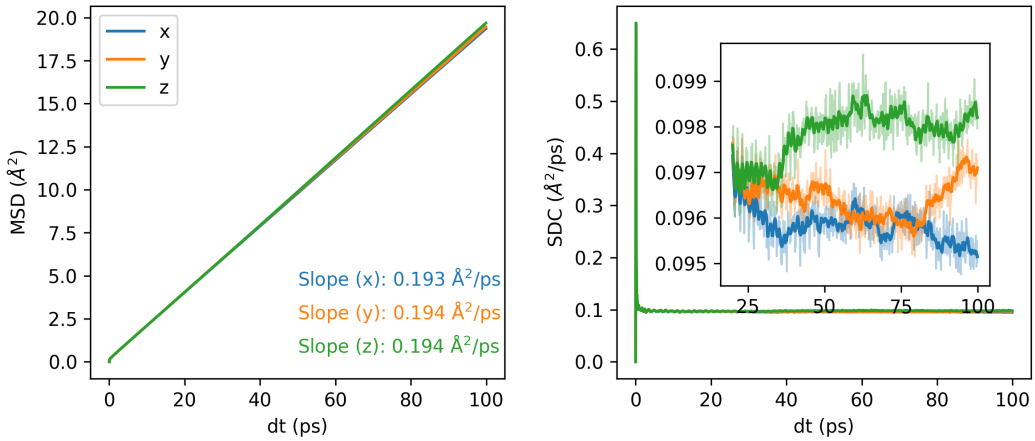
The data in this file are organized as follows:

- **column 1:** correlation time (in units of ps)
- **column 2:** MSD (in units of Å2) in the x direction
- **column 3:** MSD (in units of Å2) in the y direction
- **column 4:** MSD (in units of Å2) in the z direction
- **column 5:** SDC (in units of Å2/ps) in the x direction
- **column 6:** SDC (in units of Å2/ps) in the y direction
- **column 7:** SDC (in units of Å2/ps) in the z direction

You can use any software to visualize the data, here I recommend our [GPUMDkit](#) package, you can use the following command:

```
gpumdkit.sh -plt msd/sdc/msd_sdc
```

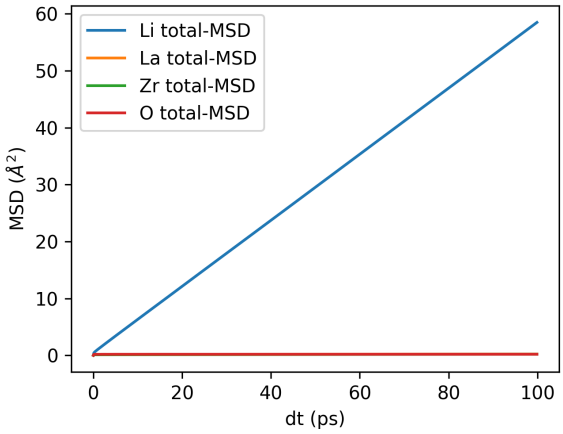
then you will see:



If you want to see the msd of all species:

```
gpumdkit.sh -plt msd_all msd.out Li La Zr O
```

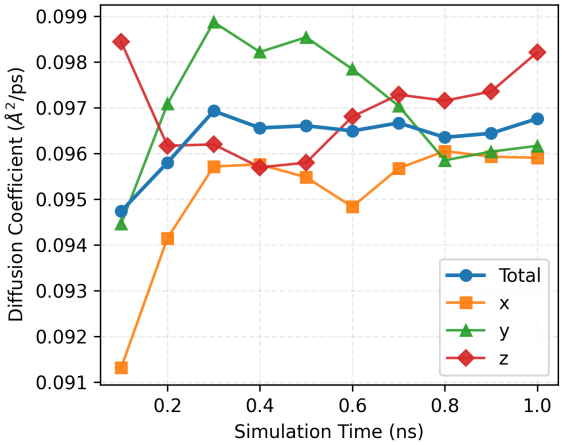
This command plots the MSD curves for all elements based on the exact group order you provided.



To ensure your total simulation time is sufficient for statistically reliable results, you can verify if the diffusion behavior has converged:

```
gpumdkit.sh -plt msd_conv
```

This command reads all the intermediate msd_step.out files and calculates the slope of the MSD for each saved block. It then plots how this slope evolves over the total simulation time.*



Step 4: Calculate Diffusion Coefficient

The **diffusion coefficient (D)** measures the rate of Li^+ ion diffusion in LLZO, a critical step toward calculating ionic conductivity. It is derived from the MSD using the Einstein relation:

$$D = \frac{1}{2N} \lim_{t \to \infty} \frac{d}{dt} \mathrm{MSD}(t)$$

where:

- N is the number of dimensions (e.g., (N = 3) for 3D diffusion in LLZO.
- $\frac{d}{dt}\mathrm{MSD}(t)$ is the slope of the MSD versus time in the diffusive regime, where MSD grows linearly.

The `msd.out` file provides the time (in ps) and MSD components (in $Å^2$). We can fit a straight line in the linear region to obtain the slope (k) (in Å²/ps) and convert the slope to the diffusion coefficient in unit of cm^2/s using:

$$D = \frac{k}{2N} \times 10^{-4}$$

Step 5: Compute Ionic Conductivity

Next, you can calculate the ionic conductivity (σ) from the diffusion coefficient (D) using the ***Nernst-Einstein*** relation:

$$\sigma = \frac{nq^2 D}{k_B T}$$

Where:

- n is the number density of ions
- q is the elementary charge
- D is the diffusion coefficient
- k_B is the Boltzmann constant
- T is the temperature

For LLZO, you need the number density (n), which depends on the number of Li^+ ions and the cell volume.

- Count the number of Li^+ ions (N_{ion}) in your `model.xyz` file. Herein, we use a $4 \times 4 \times 4$ supercell of LLZO with 3584 Li^+ ions.

- Obtain the average volume (V) (in Å^3) from the `thermo.out` file. Convert to cm^3 (Å$^3 = 10^{-24}\mathrm{cm}^3$).
- $n = \frac{N_{\mathrm{ion}}}{V}$ (in cm^{-3}).

Calculate D and σ by GPUMDkit

In fact, the `function 401` in GPUMDkit can be used to calculate σ easily. Just run `gpumdkit.sh` command and then input `401`:

```
------------>>
Input the function number:
401
>--------------------------------------------------<
| This function calls the script in calculators   |
| Script: calc_ion_conductivity.py                |
| Developer: Zihan YAN (yanzihan@westlake.edu.cn) |
>--------------------------------------------------<
Input <element> <charge> (eg. Li 1)
------------>>
Li 1
```

You only need to specify the element and charge, then you will get the following results:

```
Number of ions: 3584
Average Volume: 143800.291 Å^3
Average Temperature: 999.928 K
------------------------------
Diffusivity (D):
  D_x: 9.717e-06 cm^2/s
  D_y: 9.724e-06 cm^2/s
  D_z: 9.665e-06 cm^2/s
  D_total: 9.702e-06 cm^2/s
------------------------------
Ionic Conductivity:
  Sigma_x: 4.503e+02 mS/cm
  Sigma_y: 4.506e+02 mS/cm
  Sigma_z: 4.479e+02 mS/cm
  Sigma_total: 4.496e+02 mS/cm
------------------------------
```

The script reads the `model.xyz` and `thermo.out` files, calculating the `number of ions`, `average volume`, and `average temperature`. If these two files do not exist, you will be asked to enter these parameters manually. Finally, it will output the D and σ like the above one.

Step 6: Determining the Activation Energy

The activation energy (E_a) represents the energy barrier for Li^+ ion diffusion. It is determined by analyzing the temperature dependence of ionic conductivity using the Arrhenius equation:

$$\sigma T = A \exp\left(-\frac{E_a}{k_B T}\right)$$

where:

- σ: Ionic conductivity
- T: Temperature

- A: Pre-exponential factor
- E_a: Activation energy
- k_B: Boltzmann constant

Taking the natural logarithm:

$$\ln(\sigma T) = \ln A - \frac{E_a}{k_B T}$$

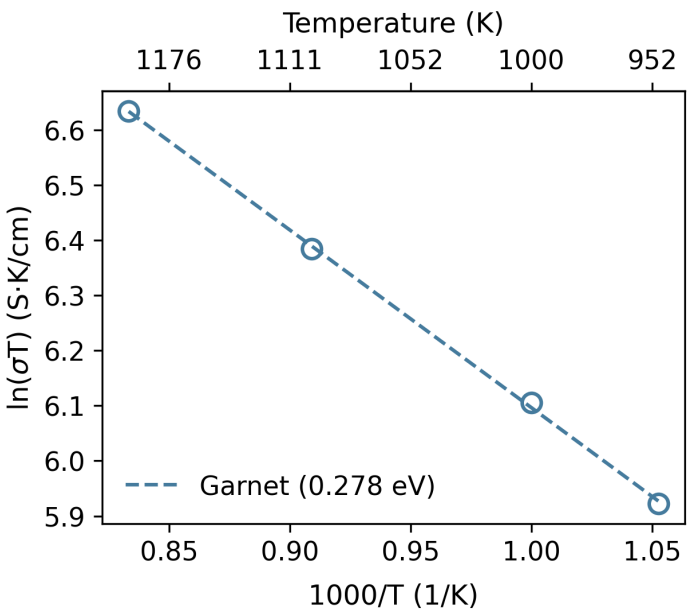
Plotting $\ln(\sigma T)$ versus $\frac{1}{T}$ yields a straight line, where the slope is $-\frac{E_a}{k_B}$. Alternatively, using $\frac{1000}{T}$ (in K^{-1}) is common, giving a slope of $-\frac{E_a}{k_B \cdot 1000}$. The activation energy is then:

$$E_a = -\text{slope} \cdot k_B \cdot 1000$$

You can also use `gpumdkit.sh -plt sigma` command, and you will get the following infomations and figure:

```
[Note] No run.in file found, assuming no replication
+-----------------------------------------------------+
|     Conductivity (unit: S/cm, Sigma*T: K S/cm)      |
+-----------------------------------------------------+
|   T (K)     |      Sigma        |      Sigma*T        |
+-----------------------------------------------------+
|    950      |    3.928e-01      |     3.731e+02       |
|    1000     |    4.484e-01      |     4.484e+02       |
|    1100     |    5.388e-01      |     5.927e+02       |
|    1200     |    6.343e-01      |     7.611e+02       |
+-----------------------------------------------------+

Garnet, Ea: 0.278 eV
at 300K, Garnet: Sigma = 8.044e-04 S/cm
Unable to display the plot due to the non-interactive backend.
The plot has been automatically saved as 'Arrhenius_sigma.png'.
```



The calculated activation energy of c-LLZO is in good agreement with the literature [2].

References

[1] Zihan Yan, Zheyong Fan and Yizhou Zhu, Improving robustness and training efficiency of machine-learned potentials by incorporating short-range empirical potentials, *J. Chem. Inf. Model.* 2026, 66, 3, 1406–1413.

[2] Zihan Yan and Yizhou Zhu, Impact of Lithium Nonstoichiometry on Ionic Diffusion in Tetragonal Garnet-Type $Li_7La_3Zr_2O_{12}$, *Chem. Mater.* 2024, 36, 23, 11551–11557.